

Prompt-History Appendix

This appendix is a transparency scaffold for the human author’s AI-assisted research-programming workflow. It should be completed before submission with the original prompt and selected follow-up prompts copied from the research log.

Original Research Prompt

The original prompt asked the assistant to audit the Rennert et al. (2022) GIVE replication archive, determine whether wildfire, biomass-burning or non-anthropogenic CO₂ emissions were already represented, design a transparent wildfire-CO₂ extension if not, run baseline and wildfire SCC experiments, diagnose double counting and produce a technical memo, figures, reproducible scripts and a manuscript.

Selected Follow-Up Prompt Themes

- RFF-SP aggregate CO₂ pathways and whether wildfire-related carbon is already embedded.
- Double-counting risk from AFOLU, LULUCF, natural-stock changes and gross fire emissions.
- Climate-driven biomass-burning projections, including boreal Canada and Russia.
- Use of residual net-persistent fire CO₂ rather than gross fire emissions as the central case.
- Deterministic stress tests, including Canada-2023-scale and high forest-fire projection scenarios.
- Sectoral marginal-damage diagnostics with saved marginal damages and sectoral values.
- Fire-flow versus atmospheric-stock percentage interpretation.
- Map revisions distinguishing source attribution from damage incidence.
- Nature-family manuscript style and movement of code-level audit details to the supplement.
- Transparent but limited disclosure of AI-assisted research programming.

Submission Note

Do not include private correspondence in this appendix unless the correspondent has granted explicit permission. The manuscript’s claim about GIVE’s exogenous emissions structure should rest on the public code audit and model documentation.